



METATRUST

Security Assessment Mapprotocol Atlas

May 22, 2026

Executive Summary

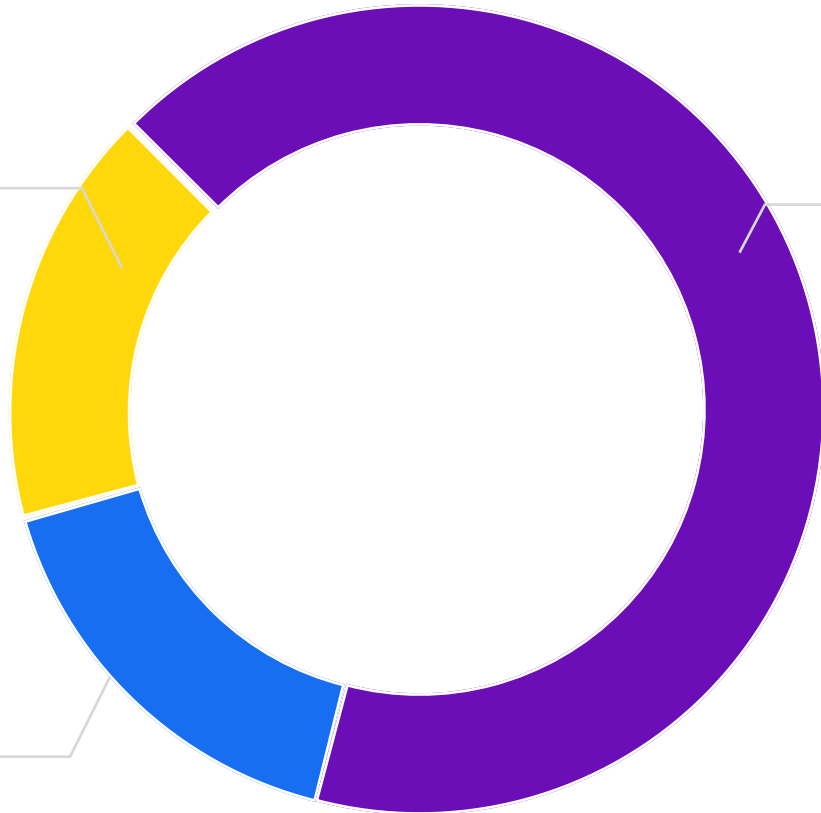
Overview			
Type	Blockchain Platform	Critical Risk Issues High Risk Issues Medium Risk Issues Low Risk Issues Informational Risk Issue	Critical Risks Are Defined As Events That Can Cause Grave Damage To A Business Operation, Result In Huge Asset Losses And Permanent Business Downtime. And It May Cause Risk Spread To Other Projects, And Endanger The Entire Ecosystem The Issue Puts A Large Number Of Users' Sensitive Information At Risk, Or Is Reasonably Likely To Lead To Catastrophic Impact For Client's Reputation Or Serious Financial Implications For Client And Users The Issue Puts A Subset Of Users' Sensitive Information At Risk, Would Be Detrimental For The Client's Reputation If Exploited, Or Is Reasonably Likely To Lead To Moderate Financial Impact. The Risk Is Relatively Small And Could Not Be Exploited On A Recurring Basis, Or Is A Risk That The Client Has Indicated Is Low-Impact In View Of The Client's Business Circumstances. The Issue Does Not Post An Immediate Risk, But Is Relevant To Security Best Practices Or Defence In Depth.
Auditors	Xue Yue		
Timeline	2026.5.15-2026.5.22		
Languages	Go, Solidity		
Methods	MetaTrust Static Analyzer And SCA Scanning Engine, Manual Audit		
Source Code	Https://Github.Com/Mapprotoocol/Atlas 6a191b70f41f47131b0a1fb17b7c5d5ae070a5f6		
Scope	Atlas/		

Total Issues	
Critical Risk Issues	0
High Risk Issues	0
Medium Risk Issues	1
Low Risk Issues	4
Informational Risk Issue	1

1 Medium Risk Issues

4 Low Risk Issues

1 Informational Risk Issue



Critical Risk Issues

High Risk Issues

Medium Risk Issues

Low Risk Issues

Informational Risk Issue

MetaTrust Audit Breakdown

MetaTrust Security Assessment Was Performed On May 22, 2026 On Project Mapprotocol With The Repository Atlas On Branch Main. The Assessment Was Carried Out By Scanning The Project's Codebase Using MetaTrust Static Analyzer And SCA Scanning Engine. There Are In Total 6 Vulnerabilities / Security Risks Discovered During The Scanning Session, Among Which 0 Critical Vulnerabilities, 0 High Risk Vulnerabilities, 1 Medium Risk Vulnerabilities, 4 Low Risk Vulnerabilities, 1 Informational Issues.

Transaction-Ordering Dependence

Timestamp Dependence

Mishandled Exceptions

Unsafe External Calls

Integer Overflow / Underflow

Number Rounding Errors

Reentrancy

Denial Of Service

Access Control

Centralization Of Power

Redundant Code

Gas Usage

Infinite Token Minting

Data Integrity Loss

Injection Type Attacks

Invalid Incoming Messages

Methodology

The Metatrust Auditing Process Follows A Routine Series Of Steps:

1. Code Review That Includes The Following
 - I. Testing The Smart Contracts Against Both Common And Uncommon Attack Vectors.
 - II. Ensuring Contract Logic Meets The Specifications And Intentions Of The Client.
 - III. Cross Referencing Contract Structure And Implementation Against Similar Smart Contracts Producedby Industry Leaders.
 - IV. Thorough Line-By-Line Manual Review Of The Entire Codebase By Industry Experts.
2. Testing And Automated Analysis That Includes The Following:
 - I. Logical Issue
 - II. Assessing The Codebase To Ensure Compliance With Current Best Practices And Industry Standards.
3. Best Practices Review, Which Is A Review Of The Smart Contracts To Improve Efficiency, Effectiveness, Clarify, Maintainability, Security, And Control Based On The Established Industry And Academic Practices, Recommendations, And Research.
4. Specific, Itemized, And Actionable Recommendations To Help You Take Steps To Secure Your Smart Contracts.

Summary of Findings

MetaTrust Security Assessment Was Performed On May 22, 2026 On Project Mapprotol With The Repository Atlas On Branch Main. The Assessment Was Carried Out By Scanning The Project's Codebase Using MetaTrust Static Analyzer And SCA Scanning Engine. There Are In Total 6 Vulnerabilities / Security Risks Discovered During The Scanning Session, Among Which 0 Critical Vulnerabilities, 0 High Risk Vulnerabilities, 1 Medium Risk Vulnerabilities, 4 Low Risk Vulnerabilities, 1 Informational Issues.

ID	Description	Severity	Status
MSA-001	Core/Vm/Evm.Go#L471 - Integer Overflow	Medium	Unfixed
MSA-002	Core/Vm/Evm.Go#L219 - Mishandled Exception	Low	Unfixed
MSA-003	Accounts/Keystore/File_cache.Go#L68 - Mishandled Exception	Low	Unfixed
MSA-004	Accounts/Abi/Unpack.Go#L165 - Code Optimization	Information	Unfixed
MSA-005	Core\Tx_pool.Go, Core\Block_validator.Go - Lack Of Malicious Information Restriction	Low	Unfixed
MSA-006	Account\ - Anonymous Measures Not Used	Low	Unfixed

Findings

 MSA-001: Core/Vm/Evm.Go#L471 - Integer Overflow Severity: Medium Checked: Integer Overflow
 Severtiy Medium Integer Overflow
 Description Unchecked Nonce Can Cause Overflow Of Nonce+1 To Zero
 File(S) Affected Core/Vm/Evm.Go#L471 <pre>// create creates a new contract using code as deployment code. func (evm *EVM) create(caller ContractRef, codeAndHash *codeAndHash, gas uint64, value *big.Int, address common.Address, typ OpCode) ([]byte, common.Address, uint64, error) { ... nonce := evm.StateDB.GetNonce(caller.Address()) //MSA-001:unchecked nonce can cause overflow of nonce+1 evm.StateDB.SetNonce(caller.Address(), nonce+1) ... }</pre>
 Recommendation Add Logic Of Check If Nonce Could Be Overflow <pre>// create creates a new contract using code as deployment code. func (evm *EVM) create(caller ContractRef, codeAndHash *codeAndHash, gas uint64, value *big.Int, address common.Address, typ OpCode) ([]byte, common.Address, uint64, error) { ... nonce := evm.StateDB.GetNonce(caller.Address()) if nonce+1 < nonce { return nil, common.Address{}, gas, ErrNonceUintOverflow } evm.StateDB.SetNonce(caller.Address(), nonce+1) ... }</pre>
 Status Acknowledged Developer: The Value Range Of Nonce Is Enough To Ensure That In The Case Of Overflow, The Transaction With Nonce Of 0 Will Not Exist In The Pending Pool, So There Will Be No Conflict, Not To Repair It

Findings

MSA-002: Core/Vm/Evm.Go#L219 - Mishandled Exception

Severity: Low

Checked: Mishandled Exception

Severtiy

Low

Mishandled Exception

Description

The Logic Of Tracer Capture Not Consider All Code Branches, This Risk Is Less Likely To Be Exploit, Whether To Modify Can Be Determined According To The Actual Situation

File(S) Affected

Core/Vm/Evm.Go#L219

```
func (evm *EVM) Call(caller ContractRef,
    addr common.Address,
    input []byte, gas uint64,
    value *big.Int) (ret []byte,
    leftOverGas uint64,
    err error)
{
    ...
    //MSA-002: the logic of tracer capture not consider all code branches
    if evm.Config.Debug && evm.depth == 0 {
        evm.Config.Tracer.CaptureStart(...)
        evm.Config.Tracer.CaptureEnd(ret, 0, 0, nil)
    }
    return nil, gas, nil
    ...
}
```

Recommendation

Add Logic Of Else Branche

```
func (evm *EVM) Call(caller ContractRef,
    addr common.Address,
    input []byte, gas uint64,
    value *big.Int) (ret []byte,
    leftOverGas uint64,
    err error)
{
    ...
    if evm.Config.Debug && evm.depth == 0 {
        evm.Config.Tracer.CaptureStart(...)
        evm.Config.Tracer.CaptureEnd(ret, 0, 0, nil)
    }
    else{
        ...
    }
    return nil, gas, nil
    ...
}
```

Status

Acknowledged

Developer: This Risk Will Not Cause Actual Security Problems, Decide Not To Repair It

Findings



MSA-003: Accounts/Keystore/File_cache.Go#L68

Severity: Low Checked: Mishandled Exception

Severtiy

Low
Mishandled Exception

Description

Did Not Check Whether Fi Is Empty, This Risk Is Less Likely To Be Exploit, Whether To Modify Can Be Determined According To The Actual Situation

File(S) Affected

Accounts/Keystore/File_cache.Go#L68

```
func (fc *fileCache) scan(keyDir string) (mapset.Set, mapset.Set, mapset.Set, error) {
    for _, fi := range files {
        ...
        //MSA-003: Did not check whether fi is empty
        modified := fi.ModTime()
        if modified.After(fc.lastMod) {
            mods.Add(path)
        }
        if modified.After(newLastMod) {
            newLastMod = modified
        }
    }
}
```

Recommendation

Add Logic Of Fi Check

```
func (fc *fileCache) scan(keyDir string) (mapset.Set, mapset.Set, mapset.Set, error) {
    for _, fi := range files {
        ...
        info, err := fi.Info()
        if err != nil {
            return nil, nil, nil, err
        }
        modified := fi.ModTime()
        if modified.After(fc.lastMod) {
            mods.Add(path)
        }
        if modified.After(newLastMod) {
            newLastMod = modified
        }
    }
}
```

Status

Acknowledged

Developer: This Risk Will Not Cause Actual Security Problems, Decide Not To Repair It

Findings



MSA-004: Accounts/Abi/Unpack.Go#L165

Severity: Information

Checked: Code Optimization

Severtiy

Information
Code Optimization

Description

Recommended To Check Whether The Err Exists First

File(S) Affected

Accounts/Abi/Unpack.Go#L165

```
func forTupleUnpack(t Type, output []byte) (interface{}, error) {
    retval := reflect.New(t.GetType()).Elem()
    virtualArgs := 0
    for index, elem := range t.TupleElems {
        marshalledValue, err := toGoType((index+virtualArgs)*32, *elem, output)
        ...
        //MSA-004: recommended to check whether the err exists first
        if err != nil {
            return nil, err
        }
        ...
    }
    ...
}
```


Recommendation

Check Whether The Err Exists First

```
func forTupleUnpack(t Type, output []byte) (interface{}, error) {
    retval := reflect.New(t.GetType()).Elem()
    virtualArgs := 0
    for index, elem := range t.TupleElems {
        marshalledValue, err := toGoType((index+virtualArgs)*32, *elem, output)
        if err != nil {
            return nil, err
        }
        ...
        ...
    }
    ...
}
```


Status

Acknowledged
Developer: This Risk Will Not Cause Actual Security Problems, Decide Not To Repair It

Findings

MSA-005: Core\Tx_pool.Go, Core\Block_validator.Go

Severity: Low Checked: Lack Of Malicious Information Filtering

Severtiy

Low

Lack Of Malicious Information Filtering

Description

Write Malicious Information In The Blockchain, Such As Virus Signatures, Politically Sensitive Topics, Etc. It Is Difficult To Delete Information After It Is Written To The Blockchain Because The Blockchain Data Cannot Be Deleted. If Malicious Information Appears In The Blockchain, It Will Be Subject To Anti-Virus Software, Political Sensitivity And Other Issues. There Is No Restriction On Sensitive Words For Data Written To The Chain In The Block Verification Related Code.

File(S) Affected

Core\Tx_pool.Go, Core\Block_validator.Go

Recommendation

Add Filtering For Malicious And Sensitive Information

Status

Acknowledge

Developer: This Risk Will Not Cause Actual Security Problems, Decide Not To Repair It

MSA-006: Account\

Severity: Low Checked: Risk Of Eavesdropping Attack

Severtiy

Low

Risk Of Eavesdropping Attack

Description

Attackers Can Use This Attack To Associate The User ID In The Blockchain With The IP Address, And In Some Cases Can Even Trace Back To The User's Home Address. Through Audit, It Is Not Found That Ring Signature, Double Key Stealth Address And Other High Anonymity Measures Are Used, So There May Be Related Risks. However, This Problem Needs To Be Comprehensively Used In Combination With Social Workers And Other Means. The Actual Utilization Success Rate Is Low, So This Problem Can Be Ignored In Combination With Actual Business

File(S) Affected

\Atlas-Main\Account

Recommendation

Ring Signature Can Be Used. Even If The Attacker Has The Private Keys Of All Members, He Cannot Find The Specific Signer, Because The Probability Of Determining The True Signer Is 1/N (N Is The Number Of The Entire Ring Members). You Can Also Use Dual Stealth Address, That Is, To Calculate An Invisible Address Into A New Unused Normal Address, So That The Two Addresses Cannot Be Contacted.

Status

Acknowledged

Developer: This Risk Will Not Cause Actual Security Problems, Decide Not To Repair It

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